IE2024 - Probability & Statistics

Time Series Analysis

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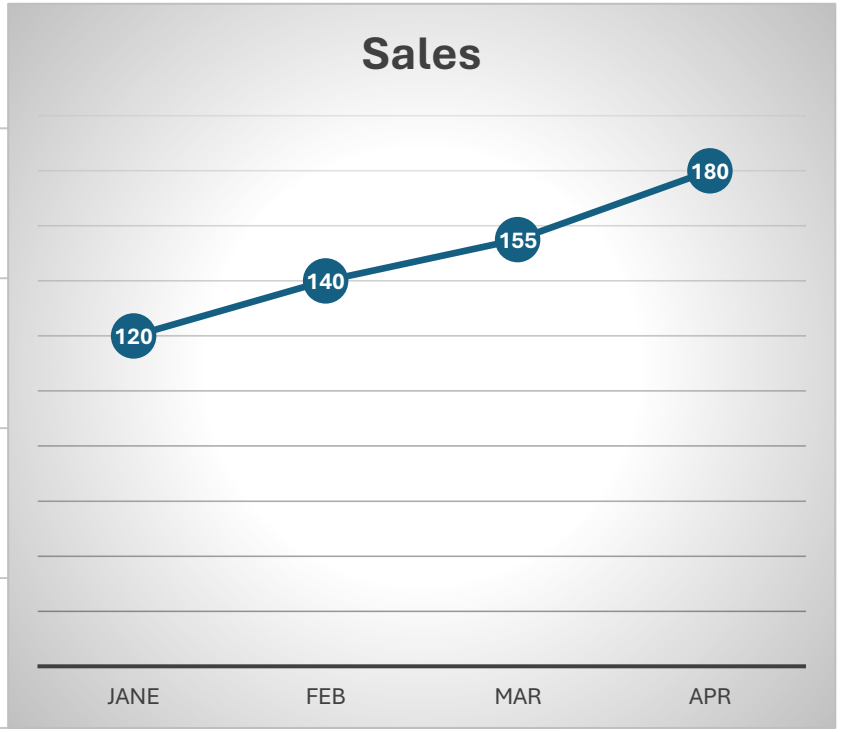


What is a Time Series?

- A **time series** is a collection of observations recorded sequentially over time. The observations are usually collected at equal time intervals such as hourly, daily, monthly, quarterly, or yearly.
- Unlike ordinary datasets, the order of observations is important because past values often influence future values.
- **Examples**
  - Daily stock prices
  - Monthly sales revenue
  - Annual rainfall
  - Hourly temperature
  - Number of website visitors per day

Example

Time	Sales
Jan	120
Feb	140
Mar	155
Apr	180



Components of a Time Series

- Trend (T)
- Seasonal Component (S)
- Cyclical Component (C)
- Irregular (Random) Component (I)

$$Time\ Series = Trend + Seasonality + Cycles + Randomness$$

Trend (T)

- Long-term movement in data.
- Example:
  - Company sales increasing every year

Seasonal Component (S)

- Regular patterns repeating within fixed periods.
- Example:
  - Ice cream sales increase every summer.
  - Retail sales increase every December.

Cyclical Component (C)

- Long-term fluctuations caused by economic or business cycles.
- Example:
  - Economic recession
  - Business expansion

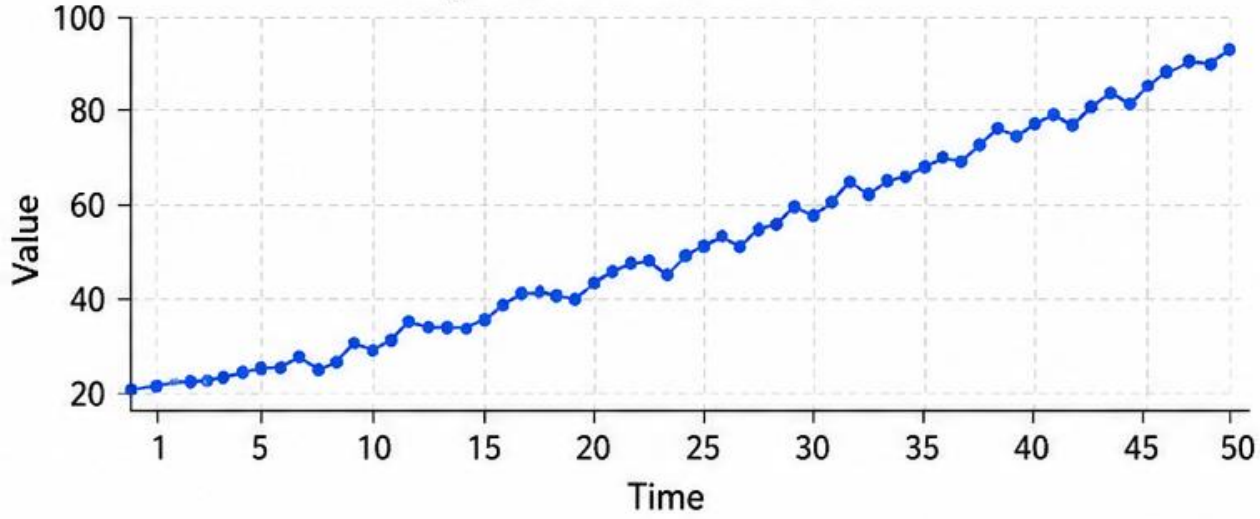
Irregular (Random) Component (I)

- Unexpected changes that cannot be predicted.
- Examples:
  - Natural disasters
  - Pandemics
  - Political events

Components of a Time Series

1. Trend (T)

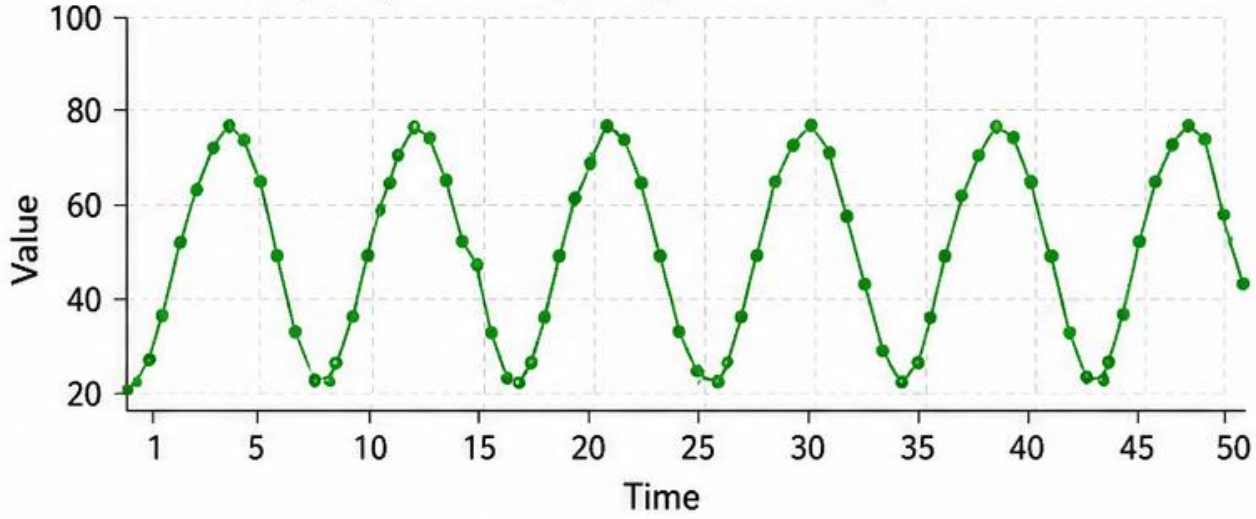
Long-term movement in data



Shows the general long-term increase or decrease in the series.

2. Seasonal Component (S)

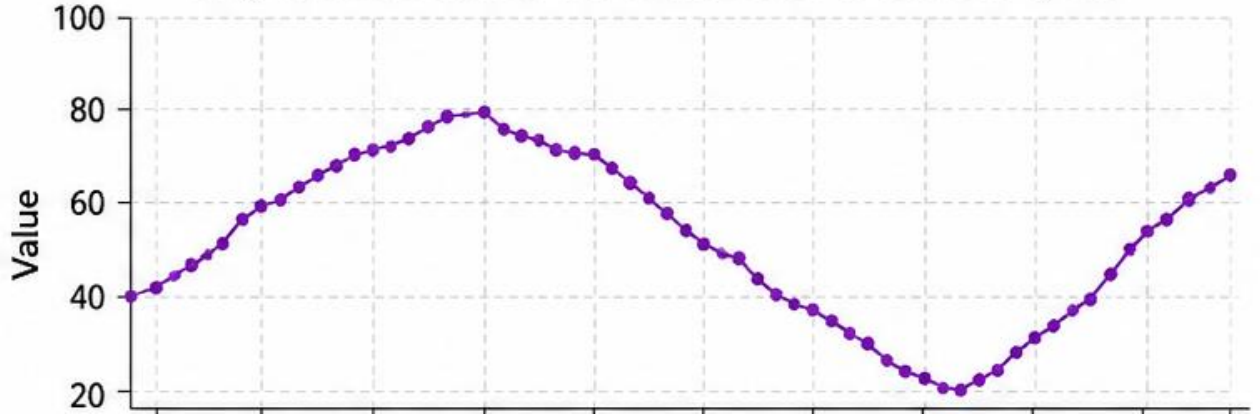
Regular patterns repeating within fixed periods



Shows regular, repeating patterns over fixed time intervals (e.g., every 12 months, every 7 days).

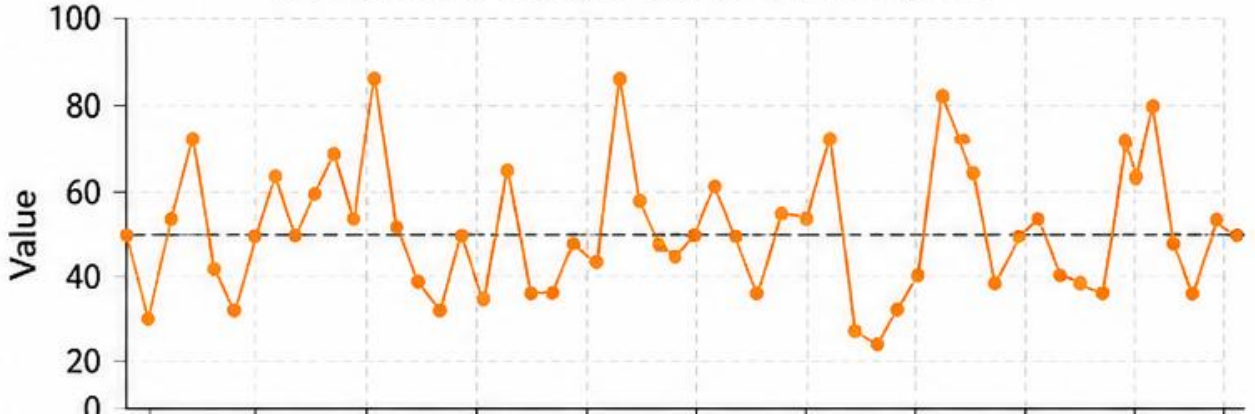
3. Cyclical Component (C)

Long-term fluctuations due to economic or business cycles



4. Irregular (Random) Component (I)

Unpredictable changes due to random factors



Additive Time Series Model

- The additive model is used when seasonal fluctuations remain approximately **constant** over time.

$$Y_t = T_t + S_t + C_t + I_t$$

- where,
  - Y_t= observed value
  - T_t= trend
  - S_t= seasonal effect
  - C_t= cyclical effect
  - I_t= irregular effect

Additive Time Series Model

- Seasonal variation remains constant.
- Difference between peak and lowest values is similar each year.
- Easy to interpret.

Multiplicative Time Series Model

- The multiplicative model is used when seasonal fluctuations **increase or decrease proportionally** with the trend.

$$Y_t = T_t \times S_t \times C_t \times I_t$$

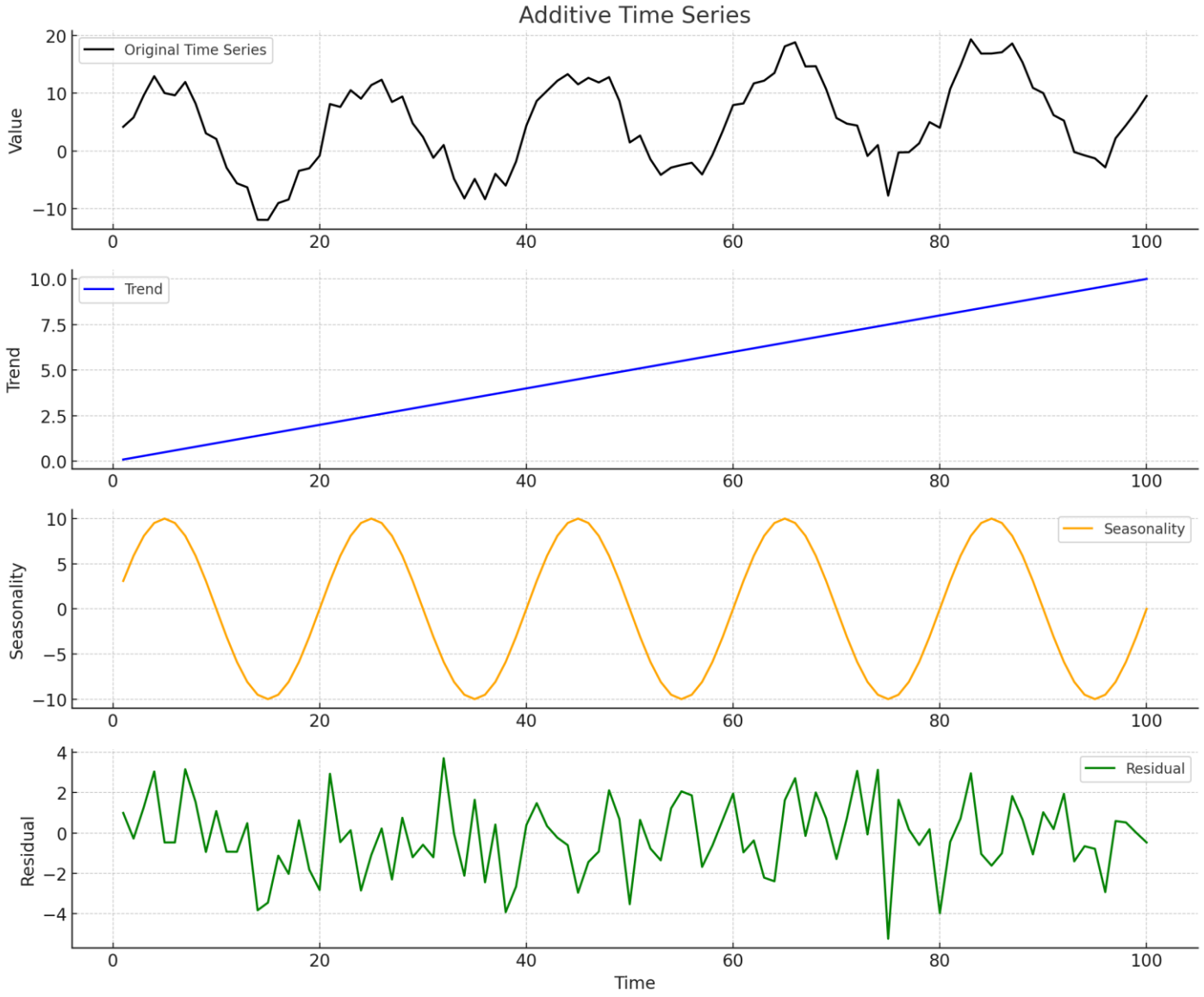
- where,
  - Y_t= observed value
  - T_t= trend
  - S_t= seasonal index
  - C_t= cyclical effect
  - I_t= irregular effect

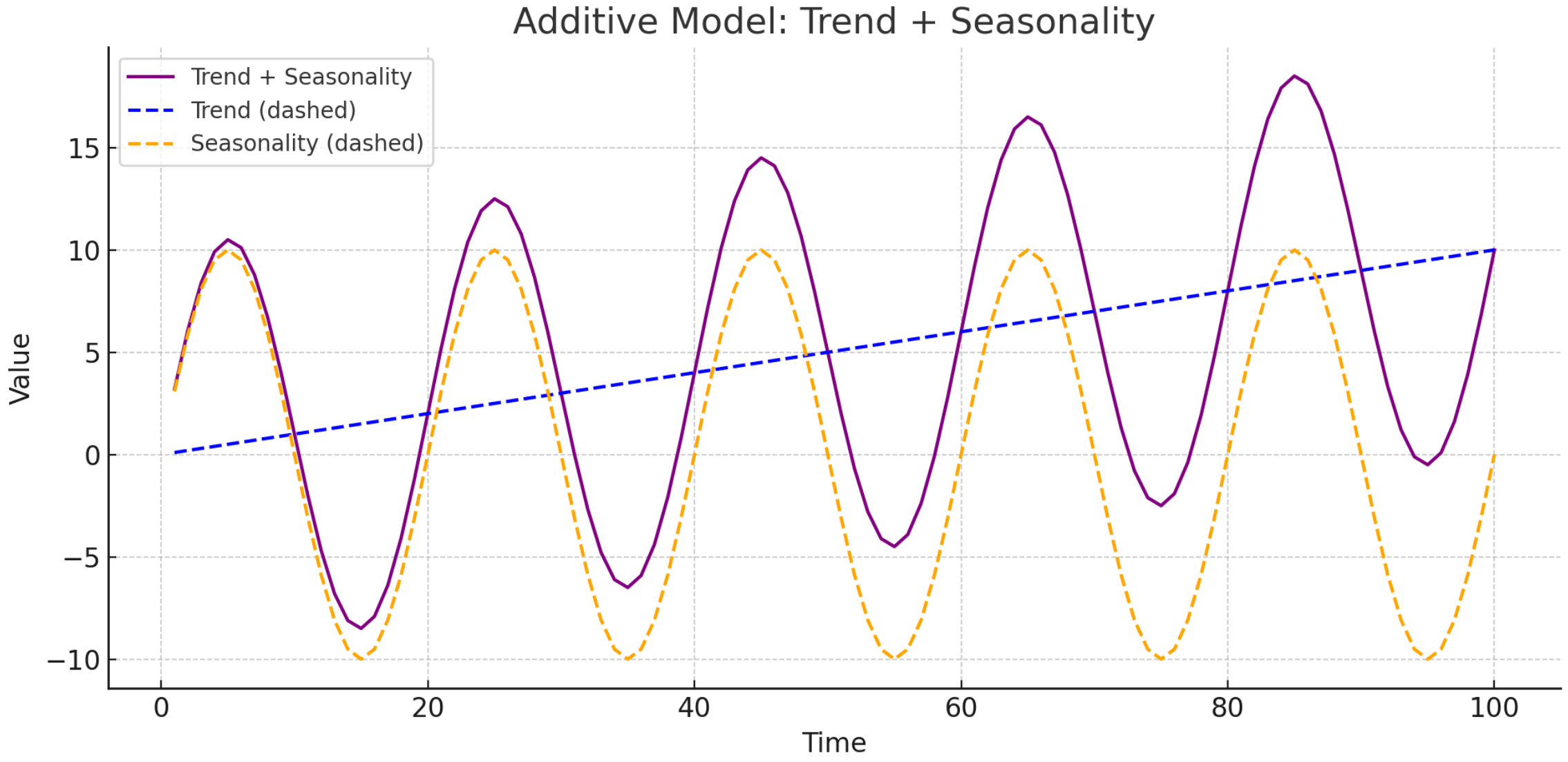
Multiplicative Time Series Model

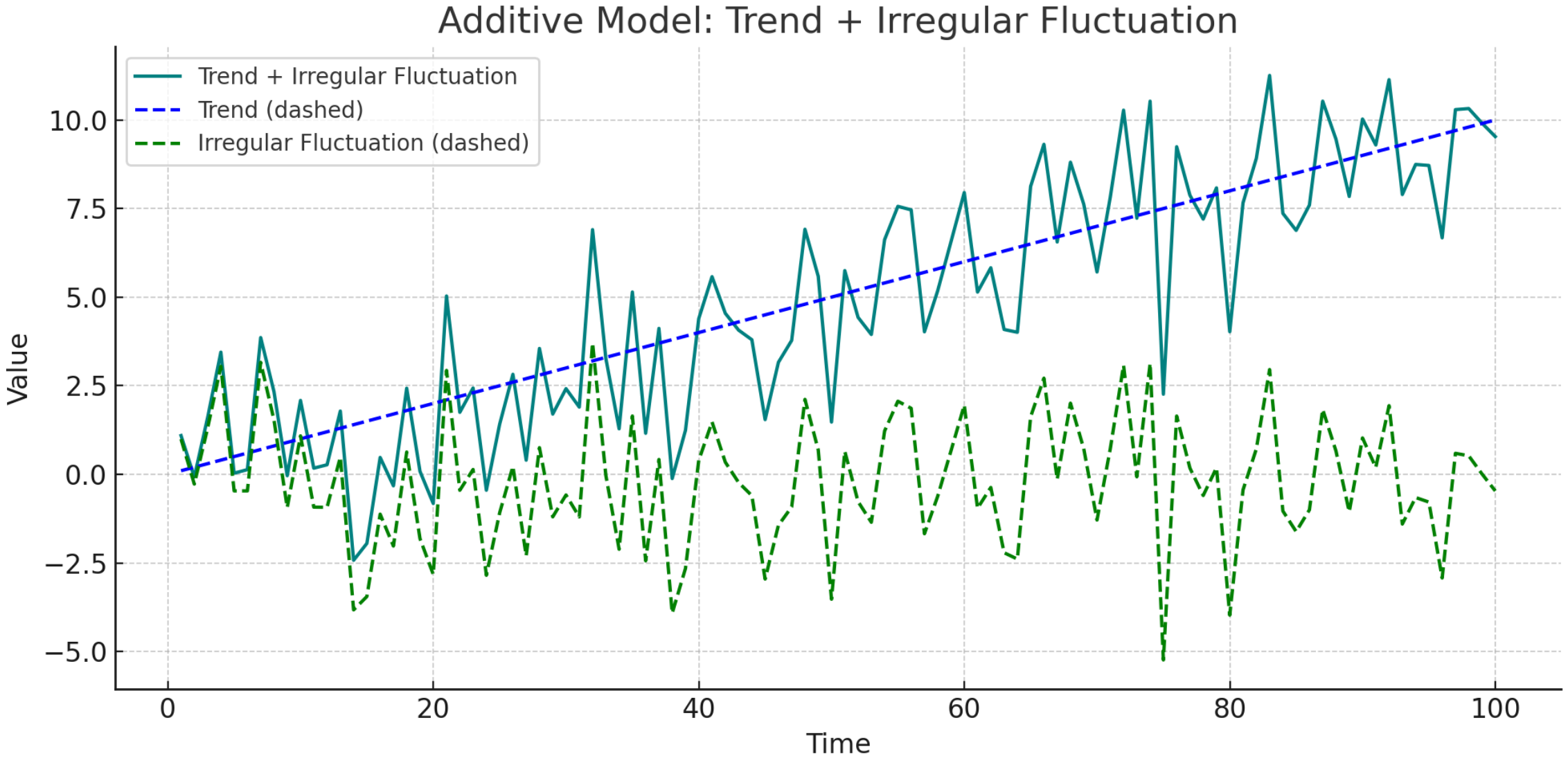
- Seasonal variation changes with the trend.
- Higher trend results in larger seasonal fluctuations.
- Widely used in business and economics.

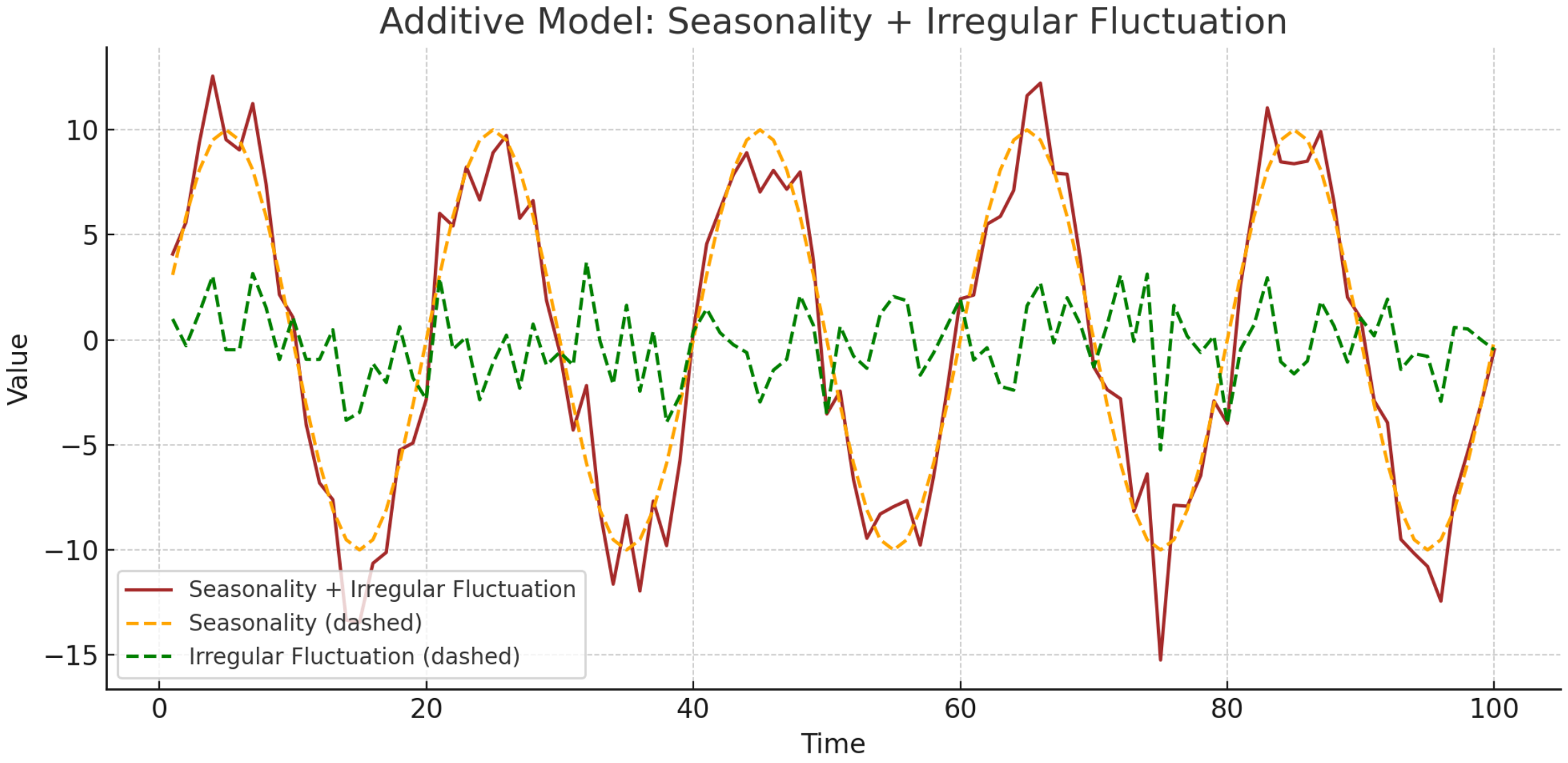
Additive VS Multiplicative

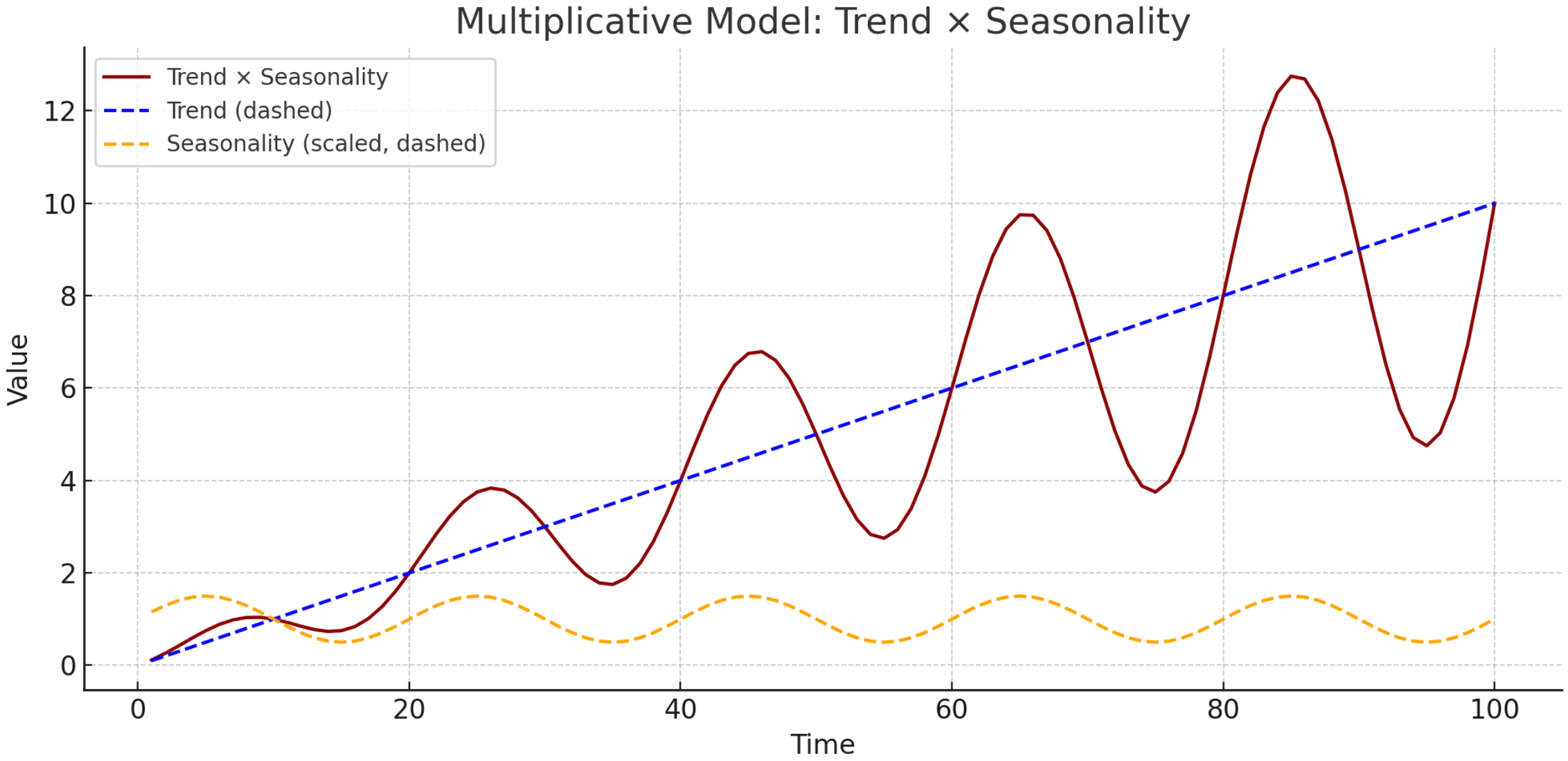
Additive Model	**Multiplicative Model**
Constant seasonal effect	Seasonal effect changes with trend
Uses addition	Uses multiplication
Constant variation	Proportional variation


Additive Time Series
Original Time Series
Value
Trend
Trend
Seasonality
Seasonality
Residual
Residual
Time


Additive Model: Trend + Seasonality
Trend + Seasonality
Trend (dashed)
Seasonality (dashed)
Value
Time
15
10
5
0
−5
−10
0
20
40
60
80
100


Additive Model: Trend + Irregular Fluctuation
Trend + Irregular Fluctuation
Trend (dashed)
Irregular Fluctuation (dashed)
Value
Time
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
0
20
40
60
80
100

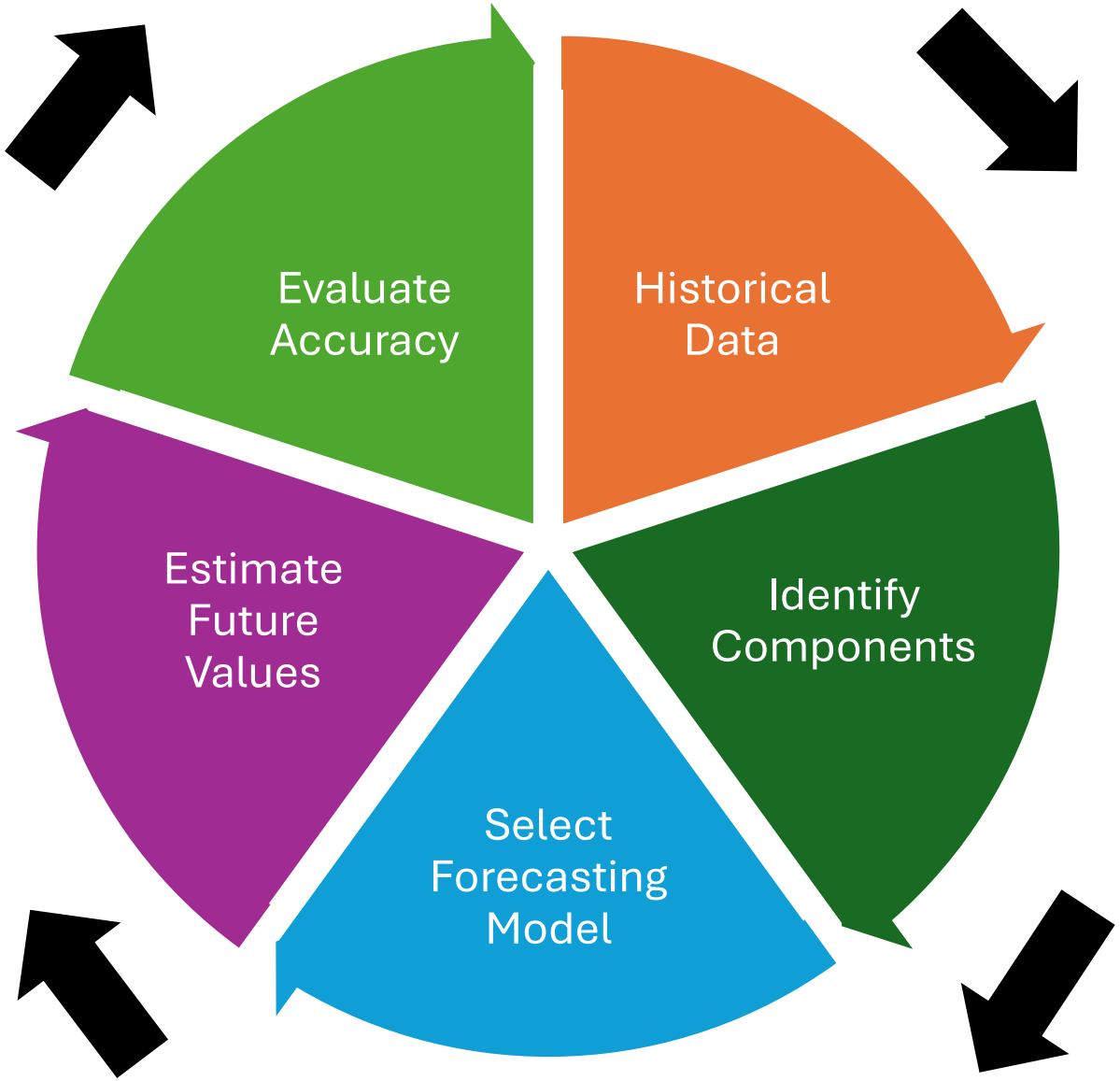

Additive Model: Seasonality + Irregular Fluctuation
Value
Time
10
5
0
−5
−10
−15
0
20
40
60
80
100
Seasonality + Irregular Fluctuation
Seasonality (dashed)
Irregular Fluctuation (dashed)


Multiplicative Model: Trend × Seasonality
Trend × Seasonality
Trend (dashed)
Seasonality (scaled, dashed)
Value
Time
0
2
4
6
8
10
12
0
20
40
60
80
100

Forecasting Using Time Series

- Forecasting is the process of predicting future values using historical observations.
- It assumes that historical patterns continue into the future.

Forecasting Using Time Series



Common Forecasting Applications

- Sales prediction
- Inventory planning
- Weather forecasting
- Electricity demand
- Population prediction

Forecasting Models

Forecasting Model	Brief Description
AR (Autoregressive) Model	Predicts future values using previous observations of the same series.
MA (Moving Average) Model	Uses previous forecast errors to improve future predictions.
ARMA Model	Combines autoregressive and moving average models for stationary time series.
ARIMA Model	Incorporates differencing with AR and MA components to model non-stationary time series.
SARIMA Model	Extends ARIMA by modelling seasonal patterns in the data.
Prophet	A forecasting model developed by Meta that automatically handles trend, seasonality, and holidays.
LSTM (Long Short-Term Memory)	A deep learning model that captures long-term dependencies in sequential time series data.